

FEASIBILITY STUDY

AI-Powered Smart Tourism Assistant Platform with Personalized Trip Planning

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230247T	Gantt Chart, Technology Stack Selection & References
230279R	Financial, Technical, Resource/Time, Risk, Social/Legal Feasibility & References
230281P	Project Overview, Objectives, Need, Existing Systems, Scope, Deliverables & References

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1. Introduction

1.1 Overview of the Project

The proposed system is an AI-Powered Smart Tourism Assistant Platform, delivered as a combined mobile and web application, that is intended to simplify the process of planning a trip. At present, travellers typically rely on a combination of separate tools and websites to plan a trip one for checking the weather, another for finding attractions, another for reviews, and often a general-purpose AI chatbot for suggestions. This proposed system brings these functions into a single platform built specifically for travel planning, rather than requiring the user to piece information together from multiple unrelated sources.

The central component of the system is an Agentic AI System that interacts with the user through a conversational, chat-based interface, similar in style to tools such as ChatGPT or Gemini. However, unlike general-purpose AI assistants, the recommendations generated by this system are not based on broad, unverified internet data. Instead, the AI operates exclusively on a curated internal database of hotels, restaurants, attractions, and local events, which is maintained and verified through an Admin Panel. This is the main distinguishing feature of the platform: recommendations are expected to be more accurate and locally relevant, since they are drawn from a dataset that has been checked rather than scraped or generated on the fly.

Beyond generating recommendations, the system is also responsible for organizing this information into a usable itinerary. Based on a traveller's stated interests, available time, and budget, the Agentic AI System generates and optimizes a personalized schedule, incorporating real-time contextual information such as current weather conditions. The platform does not process bookings directly; instead, it directs users to appropriate external platforms once a plan has been finalized, keeping the system focused on planning rather than transactions.

From a business perspective, the platform is expected to benefit two groups. Travelers gain a single, reliable point of reference for planning a trip instead of having to cross-check multiple sources. Local businesses and content providers such as hotels, restaurants, and event organizers gain a channel through which their offerings can be surfaced to travellers, with the platform planning to offer free listings initially and introducing a subscription model for continued visibility after the first year. The system is planned for deployment on AWS, allowing it to be accessed reliably across both web and mobile platforms while supporting future growth in usage.

1.2 Objectives of the Project

The objectives of this project are to:

- Design and implement a centralized smart tourism platform, delivered as both a mobile and web application, that consolidates trip planning into a single system.
- Analyse the common challenges travellers face when planning trips across multiple disconnected platforms and tools.
- Implement a personalized itinerary generation feature that adapts recommendations to a user's stated interests, time constraints, and budget.
- Develop an Agentic AI System capable of goal-oriented trip management, operating exclusively on a curated and verified internal dataset rather than general-purpose web data.
- Provide a budget planning and optimization feature that helps travellers estimate and manage trip costs.
- Establish an Admin Panel through which internal content such as hotels, restaurants, attractions, and events can be managed, verified, and kept up to date.

1.3 The Need for the Project

Trip planning today is fragmented across several unrelated platforms. A traveller planning a single trip typically needs to consult a separate weather service, a mapping tool, a review site, a booking platform, and often a general-purpose AI chatbot for suggestions, before arriving at a usable plan. This fragmentation is time-consuming and increases the likelihood that a traveller misses relevant information simply because it exists on a platform they did not think to check.

A related issue is the reliability of the information itself. General-purpose AI tools such as ChatGPT or Gemini can offer conversational travel advice, but this advice is generated from broad, general internet data rather than from a dataset that has been verified for a specific location. As a result, recommendations from these tools can be outdated, generic, or inaccurate for smaller or less known destinations, and travellers have no straightforward way to confirm which suggestions are dependable. This project addresses that gap by pairing a similar conversational interface with a dataset that is manually curated and verified through an Admin Panel, so that the reliability of a recommendation does not depend on the traveller cross-checking it elsewhere.

There is also a practical need on the planning side itself. Balancing a fixed travel timeline, a limited budget, and a set of personal interests is a non-trivial scheduling problem, and doing this manually by comparing attractions, checking distances, and estimating costs by hand is tedious even for a

single-day trip. Automating this optimization allows travellers to arrive at a workable itinerary in a fraction of the time it would otherwise take.

Together, these factors point to a genuine need for a platform that combines centralization, verified data, and automated planning in one place, rather than requiring the traveller to compensate for the shortcomings of existing tools by using several of them at once.

1.4 Overview of Existing Systems and Technologies

Several existing platforms address parts of the trip-planning problem, but none of them combine automated, personalized itinerary generation with a verified local dataset in the way the proposed system does.

Google Travel offers strong destination search and itinerary management features and integrates well with mapping and calendar tools [1]. However, its recommendations are drawn from broad, crowd-sourced, and global data rather than from a dataset that has been curated or verified for accuracy at a local level, which can make its suggestions inconsistent for smaller or less-documented destinations.

TripAdvisor is widely used for reviews and ratings of attractions, restaurants, and accommodation [2]. While useful as a reference point, it does not generate personalized itineraries automatically; the traveller must still manually piece together a schedule from the reviews and listings available.

Booking.com is primarily built around accommodation and transport reservations [3]. Itinerary planning is a secondary feature at best, and the platform does not offer any form of AI-assisted trip organization.

General-purpose AI assistants such as ChatGPT, Gemini, and Claude are increasingly used informally for travel advice, since they can hold a natural conversation and generate suggestions on demand. However, these tools generate responses from broad training data rather than a verified, destination-specific database, and they have no built-in mechanism to confirm whether a recommended hotel, restaurant, or attraction is currently accurate or still operating.

The proposed system is intended to address these individual shortcomings by combining a conversational AI interface, similar in style to the tools above, with an admin-managed, verified internal database, so that convenience and reliability are not a trade-off.

Technology Considerations

Beyond comparing existing platforms, several technology choices need to be considered for the development of this system.

For the backend, NestJS was selected for its structured, modular architecture, which is well suited to a system with multiple distinct services such as user management, AI integration, and content management [4]. For the web frontend, Next.js was chosen for its support for both server-side rendering and static content, which benefits performance on content-heavy pages such as attraction listings [5]. For the mobile application, Flutter was selected to allow a single codebase to target Android and iOS, reducing development and maintenance overhead compared to building separate native applications [6].

For data storage, PostgreSQL was chosen as a relational database, which suits the structured nature of the platform's core data like hotels, restaurants, events, and attractions and supports the relational queries needed for filtering and recommendation logic [7]. For the AI service layer, the system is planned to use Python with FastAPI and LangChain to orchestrate calls to the Gemini API, allowing the Agentic AI System to be developed and iterated on independently of the main backend [8].

Supporting services include the Google Maps API for location and mapping features [9], the OpenWeather API for real-time weather updates [10], and Firebase Cloud Messaging for push notifications on the mobile application [11].

1.5 Scope of the Project

The proposed system is designed around three primary roles, each interacting with the platform in a distinct way: the Tourist (end user), the Administrator, and the Agentic AI System, which operates as an automated component rather than a human role but is central to how the platform functions. The system is delivered across both a web application and a mobile application (Android and iOS), with certain functionalities available on one platform only, as noted below.

Tourist / User

Available on both web and mobile, the tourist will be able to:

- Register and log into the system
- Interact with the platform through a conversational, chat-based interface to plan trips
- Receive personalized itinerary recommendations based on stated interests, budget, and available time

- Search and filter attractions, restaurants, and events by district or location
- View real-time weather information relevant to their planned trip
- Access previous trips and travel history
- View nearby attractions on an interactive map (mobile only)
- Receive travel-related notifications (mobile only)

Administrator

Through the Admin Panel, the administrator will be able to:

- Add, update, and manage listings for hotels, restaurants, attractions, and local events
- Verify the accuracy of content before it becomes visible to users
- Monitor overall platform usage and user activity
- View system analytics and usage reports (web only)

Agentic AI System

Operating exclusively on data verified through the Admin Panel, the Agentic AI System is responsible for:

- Interpreting a user's travel goals and constraints from natural-language input
- Generating and optimizing personalized itineraries within the user's given timeframe
- Recommending attractions and destinations aligned with the user's stated interests
- Estimating and optimizing trip budgets
- Retrieving and presenting real-time contextual data, such as weather and local events

1.6 Deliverables

By the end of the project, the following deliverables are expected to be produced:

- A responsive, web-based tourism platform, providing trip planning, content browsing, and reporting features for both tourists and administrators.
- A cross-platform mobile application, developed for Android and iOS, offering the core trip-planning features alongside mobile-specific functionality such as an interactive nearby-attractions map and push notifications.
- A functional Agentic AI System, capable of interpreting natural-language travel requests and generating personalized, budget-optimized itineraries based on a curated internal dataset.
- An administrator dashboard, through which internal content (hotels, restaurants, attractions, and events) can be managed and verified, and through which system usage and analytics can be monitored.
- A set of RESTful API services, supporting authentication, data synchronization between the web and mobile applications, and integration with external services such as weather and mapping providers.
- A database management system, storing and organizing all internal content and user data required for the platform to function.
- Complete project documentation, including API documentation and a final project report.

2. Feasibility Study

2.1 Financial Feasibility

The Financial feasibility of the proposed AI-Powered Smart Tourism Assistant Platform (later referred this as APSTAP) is to make as that the providers will be able to list their details for free during the first year, after which a subscription fee will be required for continued listing. Users will initially have free access to the platform's planning and recommendation features. In the future, the platform may introduce advertisements specifically targeted at non-member users to generate additional revenue.

By considering on development costs; Since the proposed project is on both web application and mobile application it needs to consider mainly on deployment, maintenance and database hosting of the platform.

In deployment part cloud-based deployment has been selected to improve cost efficiency and scalable for web, mobile and AI applications. This platform is planned to deploy in Amazon Web Services (AWS) which follows a pay-as-you-go pricing model [12]. As the number of users increases and system expanded without requiring significant upfront investment in physical servers. This flexible pricing model reduces financial risk while supporting future business growth [12]. In the mobile application deployment in the initial step is planned to deploy in play store. Since publishing applications to the Apple App Store requires an annual developer subscription, the initial release of the mobile application is planned for Android through Google Play.

Another main cost aspect of the project is the database hosting. In the proposed project select to use PostgreSQL because it can be hosted through cloud services using AWS Free tier. For deployment, Amazon Relational Database Service (Amazon RDS) for PostgreSQL is selected to provide database services with automated backups and scalable storage without requiring dedicated database administration effort [7],[12].

The proposed platform needs several external services to provide intelligent travel plan suggestions and capabilities. For this it includes Gemini API for natural language understanding, Google Maps Platform for interactive maps and navigation, OpenWeather API for weather forecasting process, and Firebase Cloud Messaging for mobile notifications. Using these platforms was identified as the most financially sensible option that offer efficiency and cost effectiveness. Initially all these services are providing free usage [8], [9], [10], [11]. However, if user demand increases beyond free quotas it will need to buy those platforms in a monthly/yearly subscription basis by allowing costs to remain manageable.

The projected costs and pricing models for the required infrastructure are included in the following table

Cost Item	Project Cost
Firebase Cloud Messaging	Free tier
Google, Maps Platform,Gemini	Free tier to usage basis
AWS(production)	Usage based
Domain Name (.com)	LKR.5000/year
Play store(Android app)	25\$ registration fee

Table 1: Platform Component Cost Breakdown

2.2 Technical Feasibility

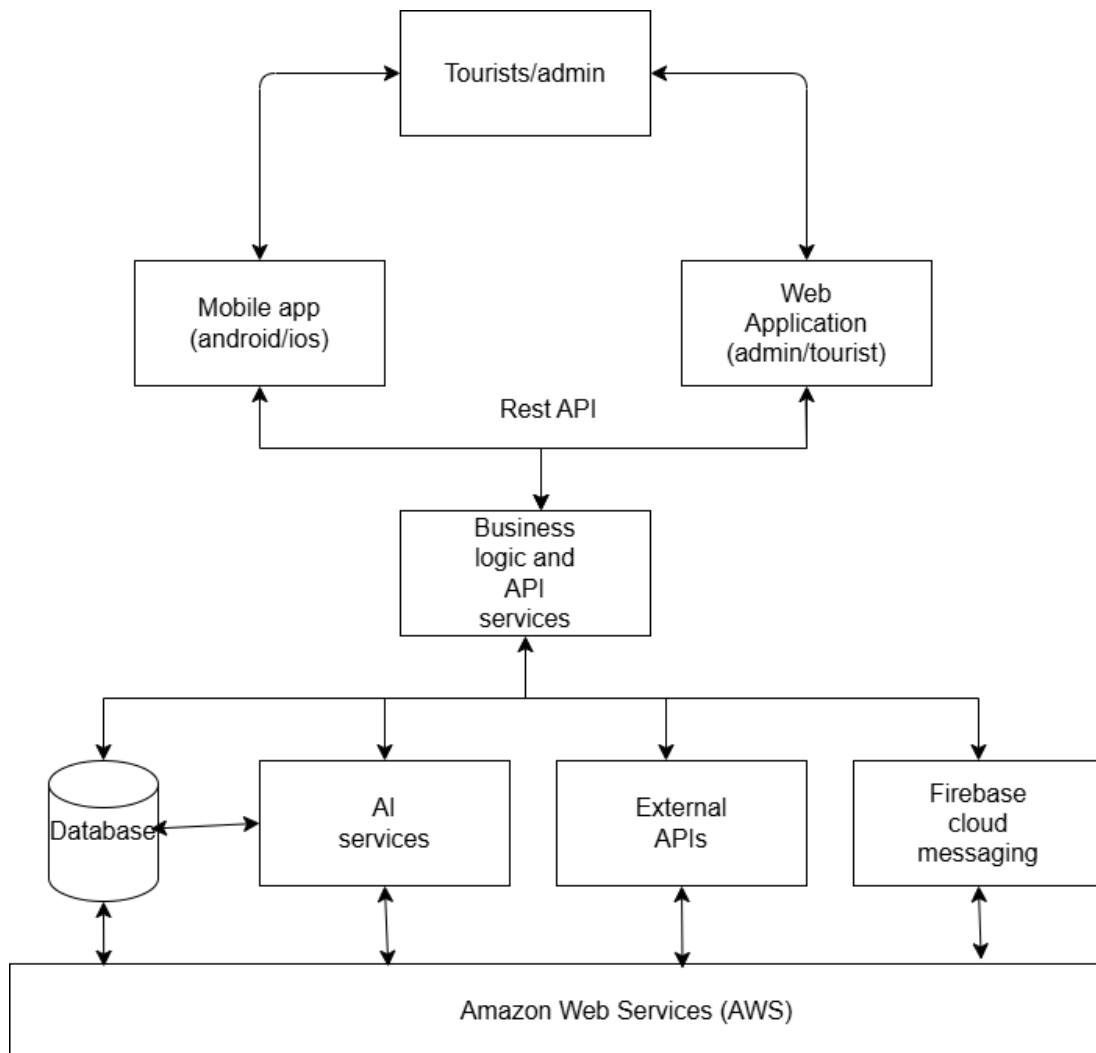


Figure 1: High-Level System Architecture of APSTAP

Technology Stack

The main technologies associated with the project are

- **Flutter** - Front-end mobile application
- **Next.js** - Frond-end web application
- **NestJS** - Back-end
- **PostgreSQL** - Database
- **Google Maps Platform**
- **OpenWeather API**
- **Firebase Cloud Messaging**
- **AWS**

Other Technologies

To support secure user authentication and authorization, the proposed system will use JSON web Tokens (JWT) for session management and role-based access control [15].

System will use bcrypt hashing algorithm for user password encryption.

Amazon Simple Email Service (Amazon SES) will be used to send email verification links during user registration, password reset, account notifications and booking details [13].

Amazon CloudWatch will be used to collect application logs, monitor system performance [12].

Git and GitHub will use for version control and collaborative software development.

Additionally, Swagger will be used to generate interactive REST API documentation and other development and testing stage documents automatically.

According to the above resources and initial analysis this project is technically feasible.

2.3 Resource and Time Feasibility

Resource Feasibility

Hardware resources: - The development crew already having personal computers with sufficient processing and memory capabilities with internet connection. And also mobile devices required for testing also available with the development crew. Therefore, the project does not have issues in hardware resources.

Software resources: - Application development will do using Visual Studio Code, version control will do using GitHub, pgAdmin will use for admin panel in PostgreSQL database, Postman will be used to test RESTful APIs and verify communication between frontend and backend. For drawing diagrams by using diagrams.io. Those Programming tools are freely available in commercial use.

The project team consist of 3 individuals with adequate potential to complete the project.

Time Feasibility

The project duration is 14 weeks. (For more details refer the project schedule document)

2.4 Risk Feasibility

AI Recommendation Accuracy Risk:

The most impactful risk is associated with the proposed platform is the accuracy and reliability of AI generated travel recommendation. Because of the system relies mostly on Large Language Models (LLM) to generate personalized travel factors some of the entities that are recommending may outdated or inaccurate. For minimization of this risk proposed system will maintain an administrator managed tourism database that provide verified information also system use previous user reviews (that can added after completing a trip/confirming bookings) and use internal database data that are added using other different travel planning websites as datasets. Furthermore, weather information will be retrieved dynamically through the OpenWeather API to ensure that itinerary recommendations consider current environmental conditions [10].

Change of policies of the freely available resources:

The proposed platform depends on some of third-party services (Gemini API, Google Maps, OpenWeather API, Firebase Cloud Messaging and Amazon services) [8]-[12]. There is a risk these may minimize user services or increase prices of packages. By reducing unnecessary cloud usage can minimize cost for it. By monitoring resource utilization using mentioned applications in technology stack (page 11).

Database Availability Risk:

Since system heavily depend on its tourism database to generate personalized recommendation, there is a high risk for database failure and data loosing. To mitigate this risk PostgreSQL will be hosted using Amazon RDS which provide automated backups, database maintenance and recovery capabilities [7],[12].

Security Risk: -

Since user sensitive data stored in the system to protect, those are hashed and encrypted before store in database. And, for user authentication system verified the user by using email-based process. [12],[15]

Health risk: -

Due to the current condition in Sri Lanaka, there is a risk of members getting infect with dengue. When this type of situation happens remaining team members should take the responsibility of the work that are belonging to infected ones and still try to complete the work in scheduled time.

2.5 Social/Legal Feasibility**Social Feasibility**

The proposed AI-Powered Smart Tourism Assistant Platform is expected to provide significant benefits to both tourists and tourism service providers by simplifying the travel planning process and improving access to reliable tourism information. Currently, travellers often rely on multiple websites and applications to search for attractions, accommodations, restaurants, weather forecasts, transportation options, and travel recommendations. This fragmented approach requires considerable time and effort to compare information from different sources. The proposed platform integrates these services into a single intelligent application [17], enabling users to generate personalized travel itineraries through a conversational interface.

Legal Feasibility

Tourism information will be reviewed and verified before it is made available to users, while AI-generated recommendations will be based on this verified information together with trusted external services such as Google Maps Platform and OpenWeather API [9], [10], while AI-assisted tourism recommendations are supported by recent research [16], [18].

Furthermore, only publicly accessible information required for the operation of the system will be collected, and copyrighted materials such as images, logos, reviews, and proprietary content will not be copied or redistributed without proper authorization. The collected data will primarily be used to initialize and verify the tourism database.

The system will also consider the requirements of the **Sri Lanka Personal Data Protection Act, No. 9 of 2022 (PDPA)** [20] by collecting only the personal information necessary to provide system functionality and processing user data for legitimate purposes. User information will not be disclosed to third parties without appropriate authorization except where required by law.

3. Considerations

Given the system's reliance on AI-generated recommendations, external service integrations, and user data, the following considerations are identified as the most significant to the platform's success.

Reliability of AI-Generated Recommendations

Since the platform's core value proposition rests on trustworthy recommendations, the consistency of the Agentic AI System's output is a central concern. This is not a hypothetical risk, general-purpose AI chatbots have been shown to frequently misrepresent smaller, local businesses specifically, since these businesses tend to have a thinner online footprint for the AI to draw from compared to larger, well-documented brands [21]. This reinforces the importance of the proposed system drawing recommendations strictly from the verified Admin Panel dataset, rather than allowing the AI to generate suggestions independently of that data, which is the core distinction between this platform and general-purpose AI tools.

Data Accuracy and Currency

Because the platform depends on an admin-managed dataset, the accuracy of that data directly determines the quality of every recommendation and itinerary generated. Listings that become outdated as an example a restaurant that has closed or an event that has been rescheduled will pose a direct risk to the reliability the platform is meant to provide. This makes a clear process for periodic review and verification of listed content an important design consideration, not just an operational afterthought.

Third-Party Service Dependency

The system's real-time features rely on several external services: the Gemini API for AI generation, the Google Maps API for location data, and the OpenWeather API for weather updates. Since these are all external dependencies, the platform's availability and performance are partly outside the development team's direct control. Rate limits, service outages, or pricing changes from any of these providers could affect the platform's reliability, and this should be factored into design decisions such as caching and fallback handling.

Security and Data Privacy

The platform stores personal user information, including travel history, preferences, and account credentials. Given that the system uses JWT-based authentication, proper token handling and secure storage of user data are necessary to prevent unauthorized access. As the platform grows, protecting this data will also be relevant to maintaining user trust in the service.

Scalability

As the number of registered users and the volume of admin-managed content grows, the system needs to maintain consistent performance, particularly for itinerary generation, which involves several data lookups and AI processing steps. Deployment on AWS is intended to support this, but scalability should also be considered at the database and API level, not only at the infrastructure level.

Usability of the Conversational Interface

Since the chat-based interface is the primary way users interact with the system, its usability is closely tied to the platform's overall success. A conversational interface that fails to understand user intent, or that returns results poorly matched to what was asked, would undermine the platform's core advantage over existing tools. Clear, well-structured responses and graceful handling of ambiguous or incomplete queries are therefore important design concerns.

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